

Numbers (8 th std)

Choose an Appropriate Answer

1. Which of the following is not an equivalent fraction of $\frac{8}{12}$?

- (a) $\frac{2}{3}$ (b) $\frac{16}{24}$ (c) $\frac{32}{60}$ (d) $\frac{24}{36}$

Ans: (b) $\frac{16}{24}$

2. The simplest form of $\frac{125}{200}$ is _____.

- (a) $\frac{25}{40}$ (b) $\frac{8}{5}$ (c) $\frac{5}{8}$ (d) None of these

Ans: (c) $\frac{5}{8}$

3. The number which is subtracted from $-\frac{6}{11}$ to get $\frac{8}{9}$ is _____.

- (a) $\frac{34}{99}$ (b) $-\frac{142}{99}$ (c) $\frac{142}{99}$ (d) $-\frac{34}{99}$

Ans: (b) $-\frac{142}{99}$

4. Which of the following pairs is equivalent?

- (a) $-\frac{20}{12}, \frac{5}{3}$ (b) $\frac{16}{-30}, -\frac{8}{15}$ (c) $-\frac{18}{36}, -\frac{20}{44}$ (d) $\frac{7}{-5}, -\frac{5}{7}$

Ans: (b) $\frac{16}{-30}, -\frac{8}{15}$

5. $-\frac{5}{4}$ is a rational number which lies between _____.

- (a) 0 and $-\frac{5}{4}$ (b) -1 and 0 (c) -1 and -2 (d) -4 and -5

Ans: (c) -1 and -2

6. Which of the following rational numbers is the greatest?

- (a) $-\frac{17}{24}$ (b) $-\frac{13}{16}$ (c) $\frac{7}{-8}$ (d) $-\frac{31}{32}$

Ans: (a) $-\frac{17}{24}$

7. The sum of the digits of the denominator in the simplest form of $\frac{112}{528}$ is _____.

- (a) 4 (b) 5 (c) 6 (d) 7

Ans: (c) 6

8. The standard form of the sum $(\frac{3}{4}) + (\frac{5}{6}) + (-\frac{7}{12})$ is _____.

- (a) 1 (b) $-\frac{1}{2}$ (c) $\frac{1}{12}$ (d) $\frac{1}{22}$

Ans: (a) 1

9. $(\frac{3}{4}) - (\frac{5}{8}) + (\frac{1}{2}) =$ _____.

- (a) $\frac{15}{64}$ (b) 1 (c) $\frac{5}{8}$ (d) $\frac{1}{16}$

Ans: (c) $\frac{5}{8}$

10. $(\frac{3}{4}) \div (\frac{5}{8}) + (\frac{1}{2}) =$ _____.

- (a) $\frac{13}{10}$ (b) $\frac{2}{3}$ (c) $\frac{3}{2}$ (d) $\frac{5}{8}$

Ans: (b) $\frac{2}{3}$

11. $(\frac{3}{4}) \times (\frac{5}{8}) \div (\frac{1}{2}) =$ _____.

- (a) $\frac{5}{8}$ (b) $\frac{2}{3}$ (c) $\frac{15}{32}$ (d) $\frac{15}{16}$

Ans: (d) $\frac{15}{16}$

12. Closure property is not true for division of rational numbers because of the number
 (a) 1 (b) -1 (c) 0 (d) $1/2$

Ans: (c) 0

13. $(1/2) - [(3/4) - (5/6)] \neq [(1/2) - (3/4)] - (5/6)$ illustrates that subtraction does not satisfy the _____ property for rational numbers.
 (a) Commutative (b) Closure (c) Distributive (d) Associative

Ans: (d) Associative

14. Which of the following illustrates the inverse property for addition?

- (a) $(1/8) - (1/8) = 0$ (b) $(1/8) + (1/8) = (1/4)$
 (c) $(1/8) + 0 = (1/8)$ (d) $(1/8) - 0 = (1/8)$

Ans: (a) $(1/8) - (1/8) = 0$

15. $(3/4) \times [(1/2) - (1/4)] = [(3/4) \times (1/2)] - [(3/4) \times (1/4)]$ illustrates that multiplication is distributive over _____

- (a) Addition (b) Subtraction (c) Multiplication (d) Division

Ans: (b) Subtraction

16. The square of 43 ends with the digit _____.

- (a) 9 (b) 6 (c) 4 (d) 3

Ans: (a) 9

17. _____ is added to 24^2 to get 25^2

- (a) 4^2 (b) 5^2 (c) 6^2 (d) 7^2

Ans: (d) 7^2

18. $\sqrt{48}$ is approximately equal to _____.

- (a) 5 (b) 6 (c) 7 (d) 8

Ans: (c) 7

19. $\sqrt{128} - \sqrt{98} + \sqrt{18} =$ _____.

- (a) $\sqrt{2}$ (b) $\sqrt{8}$ (c) $\sqrt{48}$ (d) $\sqrt{32}$

Ans: (d) $\sqrt{32}$

20. The number of digits in the square root of 123454321 is _____.

- (a) 4 (b) 5 (c) 6 (d) 7

Ans: (b) 5

21. By what number should $(-4)^{-1}$ be multiplied so that the product becomes 10^{-1} ?

- (a) $2/3$ (b) $-2/5$ (c) $5/2$ (d) $-5/2$

Ans: (b) $-2/5$

$$(-4)^{-1} = \left(-\frac{1}{4}\right)^1 = -\frac{1}{4} \Rightarrow -\frac{1}{4} \times X = \frac{1}{10} \Rightarrow X = \frac{1}{10} \times (-4) = -\frac{2}{5}$$

22. $(-2)^{-3} \times (-2)^{-2} =$ _____.

- (a) $-1/32$ (b) $1/32$ (c) 32 (d) -32

Ans: (a) $-1/32$

23. Which is not correct?

- (a) $(-1/4)^2 = 4^{-2}$ (b) $(-1/4)^2 = (1/2)^4$
(c) $(-1/4)^2 = 16^{-1}$ (d) $-(1/4)^2 = 16^{-1}$

Ans: (c) $(-1/4)^2 = 16^{-1}$

24. If $10^x / 10^{-3} = 10^9$, then x is _____.

- (a) 4 (b) 5 (c) 6 (d) 7

Ans: (c) 6

25. 0.0000000002020 in scientific form is _____.

- (a) 2.02×10^9 (b) 2.02×10^{-9} (c) 2.02×10^{-8} (d) 2.02×10^{-10}

Ans: (d) 2.02×10^{-10}

26. Which of the following can be a perfect square?

- (a) A number ending in 3 or 7 (b) A number ending in 2.
(c) A number ending with odd number of zeroes
(d) A number ending with even number of zeroes

Ans: (d) A number ending with even number of zeroes

27. Which of the following can be the square of a natural number 'n'?

- (a) Sum of the squares of first n natural numbers
(b) Sum of the first n natural numbers
(c) Sum of first (n – 1) natural numbers
(d) Sum of first 'n' odd natural numbers.

Ans: (d) Sum of first 'n' odd natural numbers.

28. Which of the following is the number non - perfect square numbers between the square of the numbers n and n + 1?

- (a) n + 1 (b) n (c) 2n (d) 2n + 1

Ans: (d) 2n + 1

29. Which of the following is the difference between the squares of two consecutive natural number is:

- (a) Sum of the two numbers (b) Difference of the numbers
(c) Twice the sum of the two numbers
(d) Twice the difference between the two numbers.

Ans: (a) Sum of the two numbers

30. Which of the following is the number of non-perfect square number between 172 and 182?

- (a) 613 (b) 35 (c) 34 (d) 70

Ans: (c) 34

31. Which of the following is the difference between the squares of 21 and 22?

- (a) 21 (b) 22 (c) 42 (d) 43

Ans: (d) 43

32. Which of the following is the number of zeroes in the square of 900?

- (a) 3 (b) 4 (c) 5 (d) 2

Ans: (b) 4

33. If M is a number such that $M \div 5$ gives a remainder of 1, then which of the following is the one's digit of M ?

- (a) 1 (b) 6 (c) 1 or 6 (d) None of these.

Ans: (b) 6

34. A number divisible by 9 is also divisible by:

- (a) 3 (b) 6 (c) 11 (d) None of these.

Ans: (a) 3

35. If $[3x74]$ is a number divisible by 9, then the least value of x is:

- (a) 1 (b) 2 (c) 3 (d) 4

Ans: (c) 3

36. If $[1X2Y6Z]$ is a number divisible by 9, then the least value of $X + Y + Z$ is:

- (a) 0 (b) 1 (c) 6 (d) 9

Ans: (d) 9

37. The number 28221 is divisible by which of the following:

- (a) 2 (b) 3 (c) 6 (d) 9

Ans: (b) 3

38. Which of the following is one's digit of a number, when divided by 5 gives a remainder of 3?

- (a) 8 (b) 3 (c) 3 or 8 (d) None of these.

Ans: (a) 8

39. If the 4-digit number $2XY7$ is exactly divisible by 3, then which of the following is the least value of $(X + Y)$?

- (a) 3 (b) 4 (c) 5 (d) 6

Ans: (a) 3

40. If a number is divisible by 2, then which of the following cannot be a one's digit in it?

- (a) 0 (b) 1 (c) 2 (d) 4

Ans: (b) 1

41. If a number is divisible by 5, then which of the following can be its one's digit?

- (a) 2 (b) 3 (c) 4 (d) 5

Ans: (d) 5

42. If a number is divisible by 10, then which of the following can be its one's digit?

- (a) 0 (b) 1 (c) 3 (d) 5

Ans: (a) 0

43. If the digit in one's place of a number is 2, then the last digit of its cube will be:

- (a) 2 (b) 4 (c) 6 (d) 8

Ans: (d) 8

44. Which of the following is correct?

- (a) Cube of a negative number is always positive.
- (b) Cube of a negative number is always negative.
- (c) Cube of a negative number may be positive or negative.
- (d) All of the above

Ans: (b) Cube of a negative number is always negative

45. If the digit in one's place of a number is 3, then the last digit of its cube will be:

- (a) 3
- (b) 6
- (c) 7
- (d) 9

Ans: (c) 7

46. If the digit in one's place of a number is 6, then the last digit of its cube will be:

- (a) 6
- (b) 3
- (c) 2
- (d) 8

Ans: (a) 6

47. The volume of a cubical box is 64 cm^3 . Which of the following is its side?

- (a) 2 cm
- (b) 4 cm
- (c) 6 cm
- (d) 8 cm

Ans: (b) 4 cm

48. Which of the following is a perfect cube?

- (a) 10000
- (b) 243
- (c) 343
- (d) 270000

Ans: (c) 343

49. If a number is doubled then which of the following is a correct statement?

- (a) Its cube is two times the cube of the given number.
- (b) Its cube is three times the cube of the given number.
- (c) Its cube is six times the cube of the given number.
- (d) Its cube is eight times the cube of the given number.

Ans: (d) Its cube is eight times the cube of the given number

50. Which of the following is equal to its own cube?

- (a) -1
- (b) -2
- (c) -3
- (d) -9

Ans: (a) -1

51. Which of the following is the cube root of 27000?

- (a) 30
- (b) 300
- (c) 3000
- (d) None of these

Ans: (a) 30

52. Which of the following is the cube root of $-64 / 243$?

- (a) $7 / 4$
- (b) $-7 / 4$
- (c) $4 / 7$
- (d) $-4 / 7$

Ans: (d) $-4 / 7$

53. What is the value of $(-1)^{-1}$?

- (a) 0
- (b) -1
- (c) 1
- (d) None of these

Ans: (b) -1

54. Which of the following is the value of 'm' in $6^m / 6^{-3} = 6^5$?

- (a) -3
- (b) -2
- (c) 3
- (d) 2

Ans: (d) 2

55. Which of the following is the standard form of 0.00001275?

- (a) 275×10^{-5} (b) 1.275×10^5 (c) 127.5×10^{-7} (d) 127.5×10^7

Ans: (a) 275×10^{-5}

56. Which of the following is used as a form of 5.05×10^6 ?

- (a) 505000 (b) 505000000 (c) 5050000 (d) 50500000

Ans: (c) 5050000

57. 1 micron = $1/1000000$ m. which of the following is its standard form?

- (a) 1.1×10^{-5} (b) 1.6×10^{-5} (c) 0.1×10^{-6} (d) 1.0×10^{-6}

Ans: (d) 1.0×10^{-6}

58. $[(1/2)^{-1} + (2/3)^2 - (3/4)^0]^{-2}$ is equal to:

- (a) $81/484$ (b) $81/169$ (c) $169/81$ (d) $16/81$

Ans: (b) $81/169$

59. Which of the following = $(100 - 99^0) \times 100$?

- (a) 10000 (b) 100 (c) 9900 (d) 99000

Ans: (c) 9900

60. What is the reciprocal of $(-3/4)^0$?

- (a) -1 (b) 1 (c) $-4/3$ (d) $4/3$

Ans: (b) 1

61. Which of the following is the value of $(4/5)^{-9} / (4/5)^{-9}$?

- (a) $(4/5)^{18}$ (b) $4/5$ (c) 1 (d) $(5/4)^9$

Ans: (c) 1

62. What should be added to $-5/4$ to get -1?

- (a) $-1/4$ (b) $1/4$ (c) 1 (d) $-3/4$

Ans: (b) $1/4$

63. What should be subtracted from $-5/4$ to get -1?

- (a) $-1/4$ (b) $1/4$ (c) 1 (d) $-3/4$

Ans: (a) $-1/4$

64. Which of the following is the identity element?

- (a) 1 (b) -1 (c) 0 (d) None of these

Ans: (c) 0

65. Which of the following is the Multiplicative identity for rational numbers?

- (a) 1 (b) -1 (c) 0 (d) None of these

Ans: (a) 1

66. Which of the following is neither appositive nor a negative rational number?

- (a) 1 (b) 0 (c) Such a rational number doesn't exist (d) None

Ans: (b) 0

67. Which of the following lies between 0 and -1?

- (a) 0 (b) -3 (c) $-2/3$ (d) $4/3$

Ans: (c) $-2/3$

68. Which of the following is the reciprocal of a ?

- (a) $-a$ (b) A (c) $1/a$ (d) $-1/a$

Ans: (c) $1/a$

69. Which of the following is the product of $7/8$ and $-4/21$?

- (a) $-1/6$ (b) $1/12$ (c) $-16/63$ (d) $-147/16$

Ans: (a) $-1/6$

70. Which of the following is the product of $(-7/8)$ and $4/21$?

- (a) $-1/6$ (b) 12 (c) $-63/16$ (d) $-16/147$

Ans: (a) $-1/6$

71. Which of the following is the reciprocal of the reciprocal of a rational number?

- (a) -1 (b) 1 (c) 0 (d) The number itself

Ans: (d) The number itself

72. 0 is not

- (a) a natural number (b) a whole number (c) an integer (d) a rational number

Ans: (a) a natural number

73. The given property $a + b = b + a$ is known as:

- (a) Commutative property (b) Distributive Property
(c) Associative Property (d) Closure property

Ans: (a) Commutative property

74. If a , b and c are whole numbers, then $a + (b + c) = (a + b) + c$. This property is called

- (a) Associative property (b) Distributive Property
(c) Commutative property (d) Closure property

Ans: (a) Associative property

75. The additive identity of any rational number is _____.

- (a) 0 (b) 1 (c) -1 (d) 2

Ans: (a) 0

76. 1 is the multiplicative identity for.....

- (a) Whole numbers (b) Integers (c) Rational numbers (d) All of the above

Ans: (d) All of the above

77. The additive inverse of 23 is

- (a) -23 (b) 32 (c) -32 (d) All of the above

Ans: (a) -23

78. How many rational numbers exist between any two distinct rational numbers?

- (a) 2 (b) 3 (c) 11 (d) Infinite number of rational numbers

Ans: (d) Infinite number of rational numbers

91. Which of the following numbers is the decimal form of $1/4$:

- (a) -0.25 (b) 0.25 (c) 2.05 (d) 2.5

Ans: (b) 0.25

92. What number should be subtracted from both the terms of the ratio 15:19 in order to make it 3:4?

- (a) 4 (b) 3 (c) 2 (d) 6

Ans: (b) 3

93. _____ is not associative for rational numbers.

- (a) Subtraction or Division (b) Addition or Multiplication
(c) Addition or Division (d) Multiplication or Division

Ans: (a) Subtraction or Division

94. $(-2)^{m+1} \times (-2)^4 = (-2)^6 \Rightarrow m =$

- (a) 0 (b) 1 (c) -1 (d) None of these

Ans: (b) 1

$$(-2)^{m+1} \times (-2)^4 = (-2)^6 \Rightarrow (-2)^{m+1+4} = (-2)^6 \Rightarrow m+5=6 \Rightarrow m=1$$

95. $(-1)^{50}$ is equal to

- (a) -1 (b) 1 (c) 50 (d) -50

Ans: (b) 1

96. $2^2 \times 2^3 \times 2^4$ is equal to:

- (a) 2^{24} (b) 2^{-5} (c) 2^9 (d) 2^{-9}

Ans: (c) 2^9

97. $3^{-2} \times 3^{-5}$ is equal to:

- (a) 3^{-7} (b) 3^{-3} (c) 3^{-10} (d) 3^7

Ans: (a) 3^{-7}

98. $5^4 / 5^2$ is equal to:

- (a) 5^6 (b) 5^{-6} (c) 5^{-2} (d) 5^2

Ans: (d) 5^2

99. The value of $(3^4)^3$ is:

- (a) 3 (b) 312 (c) 3^7 (d) None of the above

Ans: (b) 312

100. $3^2 \times 4^2$ is equal to:

- (a) 121 (b) 49 (c) 144 (d) 156

Ans: (c) 144